

NLP Milestone 1: Relation Extraction

Author: Silvia Bonomi, Samuele D’Avenia, Andrea Nugara, Maria Pronestì

November 2, 2023

1 Dataset Description

The TACRED [Zhang et al. \[2017\]](#) and TACREV [Alt et al. \[2020\]](#) datasets are investigated for this project. The second is a revisited version of the first, and further description will be provided later. The TACRED dataset is considered first. It is a large-scale relation extraction dataset, which is based on the TAC-KBP corpus of documents.

The dataset contains a set of documents where a subject, an object and a relation type are present, along with additional attributes. The relation types are selected from the TAC-KBP tasks. Moreover, in order to prevent models trained on the dataset from being biased towards predicting false positives, a high percentage of fully annotated negative instances (i.e., labeled as *no_relation*) has been included in the dataset.

Starting from this corpus of documents, the dataset contains two different groups of examples which are created differently.

- The first group is referred to as LDC examples in the reference paper. These were previously annotated by an LDC annotator as part of an annual competition. The authors believe them not to be precise enough, so a two-stage annotation task is performed using Mechanical Turk annotators.
- The second group is referred to as generated examples. The starting point is the original corpus of documents. Then, through the usage of available software to run annotations and manual annotators on Mechanical Turk, the dataset is obtained. It is also ensured that there are not too many queries regarding the same entities.

Additionally, the authors ensure that duplicate removal is performed.

The data available is divided into train, dev and test set. The division is done according to the years in which the TAC-KBP competition was run.

Now a brief description of TACREV is provided. Since data in the TACRED dataset are not publicly available, the TACREV dataset should be created following some instructions available in the GitHub repository of the paper [Alt et al. \[2020\]](#).

The TACREV dataset is a revisited version of TACRED with changes on validation and test set. The 5000 hardest examples (counted as the ones which were misclassified the most by the best models on the dataset), were relabeled by expert annotators leading to more than 50% changed annotations. This points to the data in TACRED having some potential bias. To be more precise the authors mention that there are some relations which are more frequently misclassified, with many of the location relations being wrong according to a professional annotator. This could be due to the usage of crowdsourcing instead of professional annotators.

Both datasets have a similar structure, which is described below. Dataset statistics are described

later in section 2.1.

The dataset is provided in a .conll format, with already tokenized sentences along with additional attributes. These are listed and briefly described below:

- *subj*: describes whether the token is part of the subject of the relation or not.
- *subj_type*: it contains the named entity type of the current token (if it is part of the subject).
- *obj*: describes whether the token is part of the object of the relation or not.
- *obj_type*: it contains the named entity type of the current token (if it is part of the object).
- *stanford_pos*: it contains the part of speech tag of the current token.
- *stanford_ner*: it contains the named entity tag of the current token.
- *stanford_dep*: it describes the dependency relation of the current token to its head token.
- *stanford_head*: it is a 1-based index of the dependency head of the current token.

Other than the subject and object identification, the other attributes are obtained using the Stanford Universal Dependencies Parser. The object type and subject type are part of the 23 fine-grained types used in the Stanford NER system.

2 Data Exploration and Preprocessing

This section describes the insights obtained using the code which can be found in the Jupyter Notebook, where further details can be found.

2.1 Data Exploration

A first exploration of the dataset pointed to a problem in the train data of TACRED. In fact, the last sentence provided is incomplete, and for this reason it was removed. Moreover, the TACRED train dataset available for this project is missing some of the observations. Indeed it contains 63528 documents, while the original one is reported to have 68124.

	n. sentences	avg length	min length	max length	% no rel	% no rel TACREV
train	63528	37.1	2	96	80.9%	-
dev	22631	35.5	3	95	76.0%	76.6%
test	15509	34.8	4	96	78.6%	79.9%

Table 1: Data statistics

Table 1 contains some statistics from the train, dev and test data about the sentence length and percentage of documents where there is no relation between the specified subject and object.

The train data is larger than the other two. This is expected as it contains data from more than one year, unlike the other two.

While there are some slight differences in the average length of the sentences and in the percentage of sentences with no relations, all three datasets appear to be reasonably similar for what concerns these statistics.

2.2 Data Preprocessing

The authors do not provide any description on how the text data was preprocessed, as such some insights are derived by exploring the .conll files provided.

Note that parentheses are represented using their part of speech tags, which are -LRB- and -RRB-.

It appears that text segmentation was done on standard end of sentence punctuation (such as ., !, ?, ...) or end of document, ensuring that the sentence is not split when the . represents an abbreviation. It seems that tokenization was done by selecting a token for each word including punctuation. Hence, the part of sentence "... members of parliament, causing ..." is split into 5 different tokens, namely ['members' 'of' 'parliament' ',' 'causing'].

Finally, it appears that no form of normalisation was performed on the original data, therefore it is tried out and the results are reported below with some ideas on how the usage of each technique could affect the outcome of our task.

The standard preprocessing steps were attempted. Here is a brief explanation, but it's important to note that the code and a more detailed explanation can be found within the accompanying notebook. In the initial step, sentences are reconstructed from the tokenized data contained in the JSON file using a join operation on the tokens. In the subsequent step, stopwords are removed using the list provided by the NLTK library. This operation introduces various challenges that can complicate the task and may not provide significant benefits for relation extraction. For stemming, one of the most common algorithms, the Porter stemming algorithm, is employed, while for lemmatization, the WordNetLemmatizer from the NLTK library is used. Both procedures can be considered unnecessary for the task of relation extraction because the original dataset already includes information about the grammatical role of each term. Altering the individual words in the text would require to reconstruct their grammatical roles, potentially leading to information loss. Moreover, a review of the TACRED paper reveals that, for relation extraction tasks, the architecture described relies on word representations obtained through Stanford NLP tools rather than the raw words themselves. Consequently, preprocessing the data through stemming and lemmatization may be seen as a waste of computational resources.

2.3 Using regular expressions

Given a subject and an object in a sentence, the aim of the project is to extract the correct relation between the two, by selecting one among the set of provided relations (or no relation if no match is found).

A first approach to tackle this task using only regular expressions is briefly reported below. A more detailed description is available in the Jupyter notebook.

To try and determine a possible relation between the subject and the object we try to identify some predefined patterns that could point to a specific relation. For example, if the goal is to identify whether the relation between two words is that of *per:spouse*, a possible solution is to look for sentences containing the form *[subject] ... married ... [object]* or any similar ones (where the ... are used to denote zero or more words).

Since this approach requires the development of these patterns by hand, it is quite time consuming. As such, it is used only on three relation types (and the *no_relation* type). These are *org:founded_by*, *org:member_of* and *per:spouse*.

The results are reported below using precision and recall for each of the relation types considered:

Key	Precision	Recall
<i>per:spouse</i>	0.1906	0.6250
<i>org:member_of</i>	0.0000	0.0000
<i>org:founded_by</i>	0.1047	0.3077

Table 2: Precision and Recall Values

One can immediately notice how the results vary greatly depending on the analyzed relation. This is expected, as each one requires hand-crafted patterns and some may be easier to come up with. As a matter of fact, the pattern employed for the relation *per:spouse* achieves a higher recall than the others, while the ones employed for *org:member_of* do not work. To conclude, the results obtained with this method are not satisfactory (as expected) and it may also happen that one relation is assigned two or more relation types, with no way of tie-breaking between those. Moreover, it has the big disadvantage of being time consuming and not feasible for a large number of relations.

References

- Christoph Alt, Aleksandra Gabryszak, and Leonhard Hennig. TACRED revisited: A thorough evaluation of the TACRED relation extraction task. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1558–1569, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.142. URL <https://aclanthology.org/2020.acl-main.142>.
- Yuhao Zhang, Victor Zhong, Danqi Chen, Gabor Angeli, and Christopher D. Manning. Position-aware attention and supervised data improve slot filling. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 35–45, Copenhagen, Denmark, September 2017. Association for Computational Linguistics. doi: 10.18653/v1/D17-1004. URL <https://aclanthology.org/D17-1004>.